

Unit 3: Hypothesis Testing

Calculations to conduct a one-sample T-test

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EXAMPLE — CLINIC WAITING TIME

A hospital's historical mean patient waiting time is 30 minutes. After a new check-in system is introduced, a sample of 9 patients has a mean waiting time of 28 minutes and a sample standard deviation of 3 minutes.

At the $\alpha = 0.05$ significance level, test whether the mean waiting time has **changed** from the historical value.

SOLUTION

Define the population mean difference as $\text{diff} = \mu_{\text{new}} - \mu_{\text{historical}}$.

Because we are interested in a change in **either direction**, this is a two-tailed test:

$$H_0 : \text{diff} = 0$$

$$H_A : \text{diff} \neq 0$$

Since the population standard deviation is unknown, we use the sample standard deviation and conduct a **t-test**.

The standard error of the sample mean is

$$SE = \frac{s}{\sqrt{n}} = \frac{3}{\sqrt{9}} = \frac{3}{3} = 1.$$

Therefore, the observed t -statistic is $t = \frac{\bar{x} - \mu_0}{SE} = \frac{28 - 30}{1} = -2$.

The degrees of freedom are $df = n - 1 = 9 - 1 = 8$.

For a two-tailed test with $\alpha = 0.05$, the significance level is divided equally between the two tails:

$$\frac{\alpha}{2} = \frac{0.05}{2} = 0.025.$$

For $df = 8$, the critical values are approximately

$$t^* = \pm 2.306.$$

Since $|t_{\text{obs}}| = 2 < 2.306$, the observed test statistic does not fall in the rejection region. Therefore, at the 5% significance level, we **fail to reject** H_0 .

The sample does not provide sufficient evidence to conclude that the population mean waiting time has changed from 30 minutes.